# SAGAR PUBLIC SCHOOL

**Integrated Programme • Topic: Gravitation – Level 1 (Foundation)**

# PRACTICE SHEET – 1

**Gravitation – Level 1 (Foundation): Practice Sheet 1 of 3**

| Name: _____ | Class / Section: _____ | Date: _____ |
|---|---|---|
| **Time Allowed: 60 Mins** | **Total Questions: 15** | Roll No.: _____ |

**Instructions:** Attempt all 15 questions; each has only one correct option. Sections: A – Acceleration due to Gravity, B – Gravitational Potential and Potential Energy, C – Motion of Satellite, D – Newton's Law, Gravitational Field and Kepler's Laws. Time Allowed: 60 Mins.

## Section A – Acceleration due to Gravity

**Q.1** At what altitude will the acceleration due to gravity be 25% of that at the earth's surface (given radius of earth is $R$)?

(A) $R/4$ (B) $R$
(C) $3R/8$ (D) $R/2$

[Motion, Ex-I Q.1]

**Q.2** Assuming the earth to be a sphere of uniform mass density, a body weighed 300 N on the surface of earth. How much it would weigh at $R/4$ depth under surface of earth?

(A) 75 N (B) 375 N
(C) 300 N (D) 225 N

[JEE Main 2024, 6th April Evening Shift]

**Q.3** A 90 kg body placed at $2R$ distance from surface of earth experiences gravitational pull of: ($R$ = Radius of earth, $g = 10$ m s$^{-2}$)

(A) 300 N (B) 225 N
(C) 100 N (D) 120 N

[JEE Main 2024, 4th April Evening Shift]

**Q.4** If the radius of the earth be increased by a factor of 5, by what factor its density be changed to keep the value of $g$ the same?

(A) 1/25 (B) 1/5
(C) $1/\sqrt{5}$ (D) 5

[Motion, Ex-I Q.4]

## Section B – Gravitational Potential and Potential Energy

**Q.5** To project a body of mass $m$ from earth's surface to infinity, the required kinetic energy is (assume, the radius of earth is $R_E$, $g$ = acceleration due to gravity on the surface of earth):

(A) $\frac{1}{2}mgR_E$ (B) $4mgR_E$
(C) $mgR_E$ (D) $2mgR_E$

[JEE Main 2024, 6th April Morning Shift]

**Q.6** A hollow spherical shell is compressed to half its radius. The gravitational potential at the centre

(A) increases (B) decreases
(C) remains same (D) during the compression increases then returns at the previous value

[Motion, Ex-I Q.11]

**Q.7** Earth has mass 8 times and radius 2 times that of a planet. If the escape velocity from the earth is 11.2 km/s, the escape velocity in km/s from the planet will be:

(A) 8.4 (B) 11.2
(C) 5.6 (D) 2.8

[JEE Main 2025, 28th January Evening Shift]

**Q.8** The escape velocity for a planet is $v_e$. A tunnel is dug along a diameter of the planet and a small body is dropped into it at the surface. When the body reaches the centre of the planet, its speed will be

(A) $v_e$ (B) $\frac{v_e}{\sqrt{2}}$ (C) $\frac{v_e}{2}$ (D) zero

[Motion, Ex-I Q.10]

## Section C – Motion of Satellite

**Q.9** A satellite of the earth is revolving in circular orbit with a uniform velocity $V$. If the gravitational force suddenly disappears, the satellite will

(A) continue to move with the same velocity in the same orbit
(B) move tangentially to the original orbit with velocity $V$
(C) fall down with increasing velocity
(D) come to a stop somewhere in its original orbit

[Motion, Ex-I Q.17]

**Q.10** A satellite revolves in the geostationary orbit but in a direction east to west. The time interval between its successive passing about a point on the equator is

(A) 48 hrs (B) 24 hrs
(C) 12 hrs (D) never

[Motion, Ex-I Q.18]

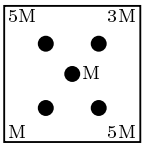
**Q.11** If a satellite orbiting the Earth is 9 times closer to the Earth than the Moon, what is the time period of rotation of the satellite? Given rotational time period of Moon = 27 days and gravitational attraction between the satellite and the moon is neglected.

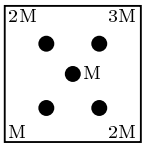
(A) 3 days (B) 27 days
(C) 81 days (D) 1 day

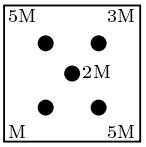
[JEE Main 2025, 23rd January Evening Shift]

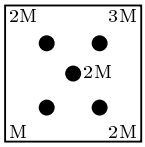
## Section D – Newton's Law, Gravitational Field and Kepler's Laws

**Q.12** A mass is at the center of a square, with four masses at the corners as shown.

(A) 


(B) 


(C) 


(D) 


Rank the choices according to the magnitude of the gravitational force on the center mass.

(A) $F_A = F_B < F_C = F_D$

(B) $F_A > F_B < F_D < F_C$

(C) $F_A = F_B > F_C = F_D$

(D) none

[Motion, Ex-I Q.16]

**Q.13** A planet of mass $m$ is in an elliptical orbit about the sun ($m \ll M_{sun}$) with an orbital period $T$. If $A$ be the area of orbit, then its angular momentum would be

(A) $\frac{2mA}{T}$

(B) $mAT$

(C) $\frac{mA}{2T}$

(D) $2mAT$

[Motion, Ex-I Q.26]

**Q.14** A planet ($P_1$) is moving around the star of mass $2M$ in the orbit of radius $R$. Another planet ($P_2$) is moving around another star of mass $4M$ in a orbit of radius $2R$. Ratio of time periods of revolution of $P_2$ and $P_1$ is _____.

(A) $\frac{1}{2}$

(B) 2

(C) 4

(D) $\frac{1}{4}$

[JEE Main 2026, 2nd April Morning Shift]

**Q.15** A light planet is revolving around a massive star in a circular orbit of radius $R$ with a period of revolution $T$. If the force of attraction between planet and star is proportional to $R^{-3/2}$ then choose the correct option:

(A) $T^2 \propto R^{7/2}$

(B) $T^2 \propto R^3$

(C) $T^2 \propto R^{5/2}$

(D) $T^2 \propto R^{3/2}$

[JEE Main 2024, 1st February Evening Shift]

# ANSWER KEY – Practice Sheet 1

Answers are taken from the source answer keys: Motion (Answer Ex-I) and the JEE PYQ file (Answer Key at the end).

**1.** [Motion Ex-I Q.1] B

**2.** [JEE PYQ file Q.22] D

**3.** [JEE PYQ file Q.26] C

**4.** [Motion Ex-I Q.4] B

**5.** [JEE PYQ file Q.23] C

**6.** [Motion Ex-I Q.11] B

**7.** [JEE PYQ file Q.14] C

**8.** [Motion Ex-I Q.10] B

**9.** [Motion Ex-I Q.17] B

**10.** [Motion Ex-I Q.18] C

**11.** [JEE PYQ file Q.16] D

**12.** [Motion Ex-I Q.16] A

**13.** [Motion Ex-I Q.26] A

**14.** [JEE PYQ file Q.5] B

**15.** [JEE PYQ file Q.29] C

# SAGAR PUBLIC SCHOOL

**Integrated Programme • Topic: Gravitation – Level 1 (Foundation)**

## SELF ASSESSMENT – Practice Sheet 1

*Fill this in after checking your answers with the Answer Key.*

| | |
|---|---|
| **Time Allowed:** 60 Mins | **Time Taken:** _____ Mins |
| Start time: _____ | End time: _____ |
| Finished within the allowed time? ☐ Yes ☐ No | (extra / saved time: _____ Mins) |

| Total | Attempted | Correct | Incorrect | Not attempted |
|---|---|---|---|---|
| 15 | | | | |

Score (Correct / 15): _____ Percentage: _____ %

**Question-wise tracker** write ✓ (correct), ✗ (incorrect) or – (not attempted)

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|
| | | | | | | | |
| 9 | 10 | 11 | 12 | 13 | 14 | 15 | |
| | | | | | | | |

**Reason for the incorrect / unattempted questions** (write question numbers)

| | |
|---|---|
| Concept not clear: _____ | Formula forgotten: _____ |
| Calculation / silly mistake: _____ | Ran out of time: _____ |

**Personal feedback guide (tick the band that matches your score)**

| | Correct | What to do next |
|---|---|---|
| ☐ | 14–15 | Excellent. Concept and speed are both strong. Move to harder JEE-level problems and keep revising formulae once a week. |
| ☐ | 11–13 | Good. Re-do every incorrect question without looking at the solution, then note the formula or idea you missed. |
| ☐ | 7–10 | Fair. Revise the theory of escape speed, orbital speed, energies of a satellite and Kepler's laws, then re-attempt this sheet after 3 days. |
| ☐ | 0–6 | Needs attention. Re-read the chapter notes and solved examples first, then attempt this sheet again in two parts of 30 mins. |

**My personal feedback / action plan:**

_____

_____

_____

_____

**Teacher's remarks:**

_____

_____

Student's signature: _____ Teacher's signature: _____